COMPUTATIONAL APPROACHES TO LEXICAL SEMANTIC CHANGE DETECTION WITH WIC-BASED AND LLM-BASED SENSE INDUCTION

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Lexical Semantic Change Detection

- **Binary Task**: Given a set of target words, determine which words have gained or lost senses between C1 and C2, and which have remained semantically stable.

- **Ranking task**: Rank a set of target words according to the degree of lexical semantic change between C1 and C2 by comparing their sense frequency distributions across the two corpora.

Lexical Semantic Change Detection

- **COMPARE:** Given two usages of a target word, this task focuses on estimating semantic change by considering pairs of usages in which one occurrence comes from the earlier corpus and the other from the later one.

COMPARE task

Introduction

- language is constantly evolving, reflecting processes within both language and society

cell $\longrightarrow$ small monastery cell, prison room

- later, it gained a new sense (telecommunications) in 1977

(Online Etymology Dictionary)

Thesis Roadmap

1- Introduction

2 - Research Questions

3 - LSCD Spanish dataset

4 - WSI optimization for LSCD

5 - Enhancing LLMs for LSCD

6 - Conclusions

LSCD - Applications

Linguistics: reveals how social and cultural dynamics, cognitive processes,and lexical structure interact to shape word usage over time

Lexicography: tracks the emergence of new word senses and helps refine historical dictionaries

LSCD - Applications

Social and Political Science: investigates cultural and ideological shifts, the evolution of political discourse, and semantic signals tied to societal phenomena such as armed conflict

Digital Humanities: enables computational analysis of historical corpora across disciplines

Research question 1

Does optimization for WSI transfer to LSCD?

DURel Relatedness Scale

4: Identical

3: Closely Related

2: Distantly Related

1: Unrelated

(Schlechtweg et al. 2018)

Example

- Usage A (1850): The prisoner was confined to a narrow cell in the tower.

- Usage B (2001): She forget her cell at home and missed the call.

- Judgment: 1 - Unrelated

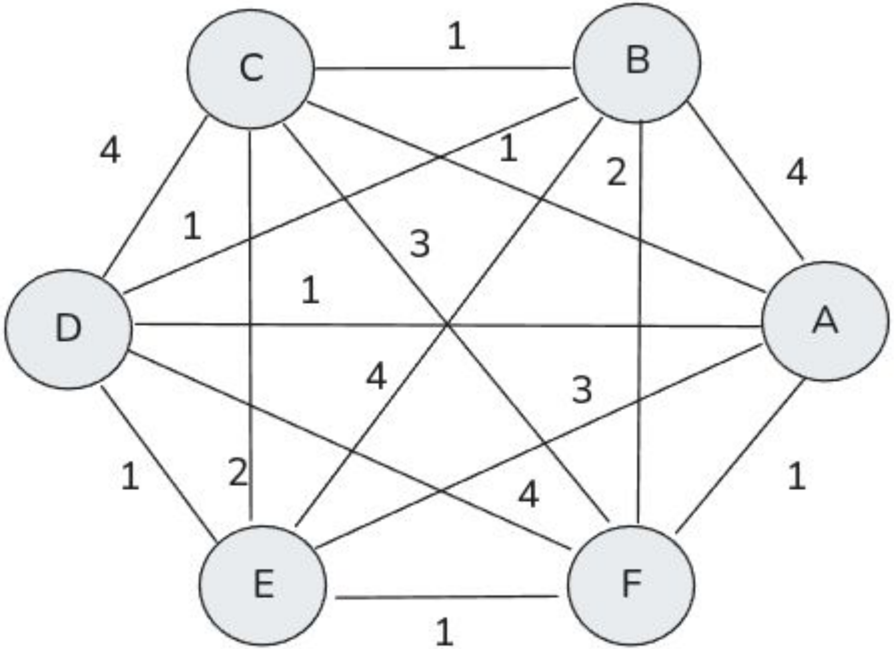
Word Usage Graph (WUG)

- $G = (U, E, W)$, weighted , undirected graph
- $u \in U$, represents word usages
- $w \in W$, represents semantic proximity of pair of usages $(u1,u2) \in E$

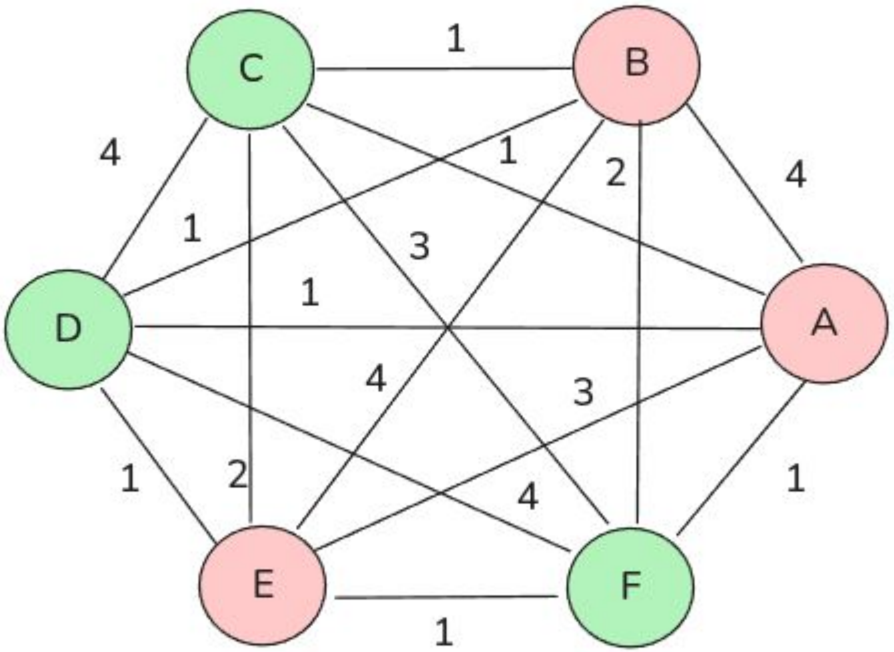
Example

ID	Period	Usage	Meaning
A	1810	Fui al banco a retirar dinero.	financial institution
B	1855	El banco me concedió un préstamo.	financial institution
E	1906	El banco central emitió un comunicado oficial.	financial institution
C	1994	Nos sentamos en el banco del parque a descansar.	bench
D	2008	El banco estaba mojado después de la lluvia.	bench
F	2020	Se quedó dormido en un banco de madera junto al río.	wooden bench

WUG



WUG



Background - SemEval 2020 Task 1

- 4 datasets presented and a set of target words W={w1,w2,…,wn}
- 2 tasks
  - binary task
  - ranking task
-

Example

Period Usage

A (1810) Fui al banco a retirar dinero. (financial institution)

B (1855) El banco me concedió un préstamo. (financial institution)

E (1906) El banco central emitió un comunicado oficial. (financial institution)

C (1994) Nos sentamos en el banco del parque a descansar. (bench)

D (2008) El banco estaba mojado después de la lluvia. (bench)

F (2020) Se quedó dormido en un banco de madera junto al río. (wooden bench)

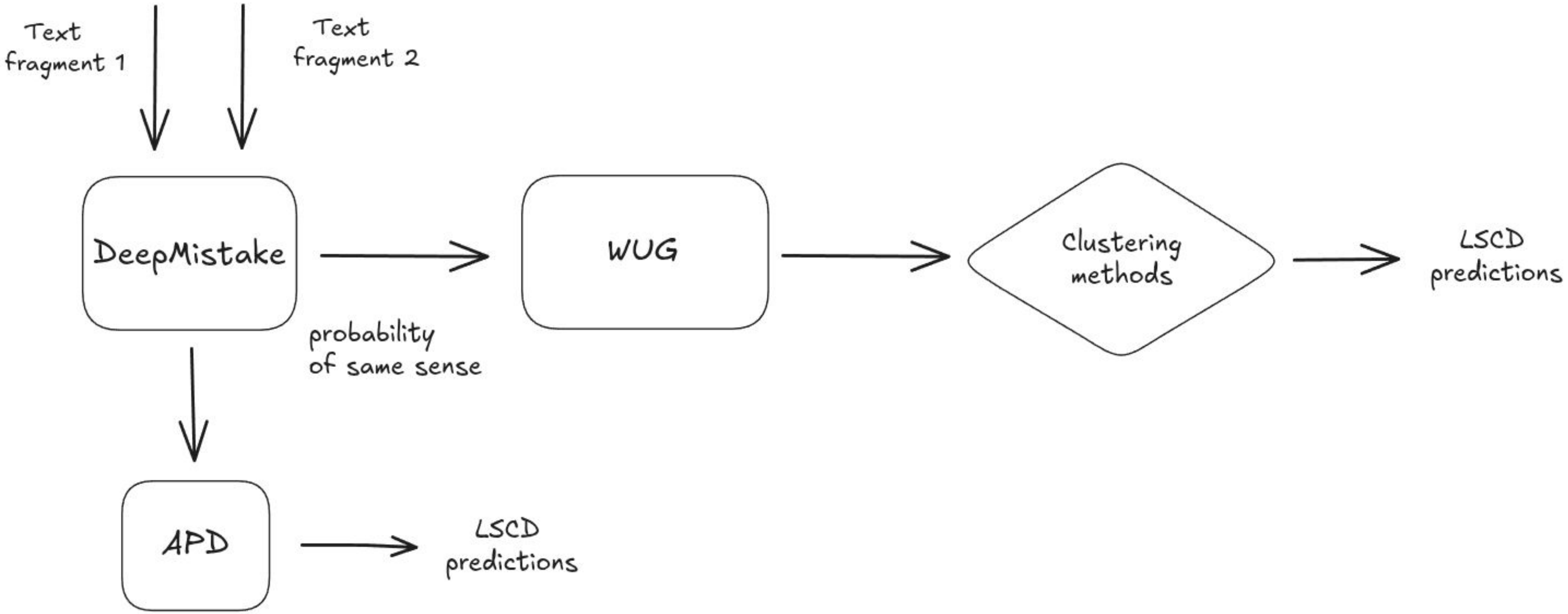
Hypothesis

Optimizing clustering configurations for WSI will improve their downstream performance on LSCD.

Datasets

Dataset	Id	n	C1	C2
German	DWUG DE	24	1800 - 1899	1946 - 1990
Spanish	DWUG ES	60	1810 - 1906	1994 - 2020
English	DWUG EN	37	1810 - 1860	1960 - 2010

Experimental setup



WUG

- complete graph generated -> [0.0 - 1.0]
    - each language

- hyperparameters
    - threshold t (edges below t are cut them out)
    - binarize (weighted edges -> [0, 1])
    - normalized (-> [0, 1])
    - use disconnected edges (-> [True, False])
    - quantile (-> [1,..., 10])
    - fill_diagonal (-> [True, False])

Clustering methods

Clustering method	**Parameter**	**Values**
Chinese Whispers	weighting	[lin, top, log]
Correlation Clustering	threshold_cc	0.5
WSBM	distribution	[binomial, poisson, discrete, normal, exponential]
Spectral Clustering	number of clusters	[Silhouette, Calinski-Harabas, Eigengap]

German results

Clustering method	Optimized - WSI		Optimized - LSCD	
	ARI	Spearman	ARI	Spearman
Chinese Whispers	.622 ± .130	.768 ± .241	.602 ± .078	.768 ± .241
Correlation clustering	.622 ± .132	.873 ± .110	.666 ± .096	.715 ± .207
WSBM	.710 ± .068	.911 ± .156	.613 ± .158	.828 ± .169
Spectral Clustering - Silhouette	.771 ± .090	.873 ± .141	.715 ± .067	.835 ± .130
Spectral Clustering - Calinkski - Harabasz	.755 ± .087	.834 ± .077	.702 ± .087	.774 ± .111
Spectral Clustering - Eigengap	.633 ± .144	.807 ± .202	.426 ± .208	.596 ± .213

Spanish results

Clustering method	Optimized - WSI		Optimized - LSCD	
	ARI	Spearman	ARI	Spearman
Chinese Whispers	.475 ± .049	.590 ± .260	.453 ± .066	.498 ± .254
Correlation clustering	.412 ± .113	.661 ± .106	.411 ± .115	.615 ± .121
WSBM	.444 ± .022	.553 ± .186	.480 ± .078	.602 ± .243
Spectral Clustering - Silhouette	.351 ± .108	.578 ± .173	.371 ± .112	.420 ± .187
Spectral Clustering - Calinkski - Harabasz	.363 ± .097	.420 ± .239	.316 ± .113	.301 ± .271
Spectral Clustering - Eigengap	.310 ± .084	.595 ± .152	.301 ± .094	.567 ± .189

English results

Clustering method	Optimized - WSI		Optimized - LSCD	
	ARI	Spearman	ARI	Spearman
Chinese Whispers	.243 ± .098	.700 ± .209	.256 ± .129	.598 ± .198
Correlation clustering	.192 ± .104	.307 ± .194	.080 ± .076	.293 ± .189
WSBM	.226 ± .144	.562 ± .297	.045 ± .025	.555 ± .214
Spectral Clustering - Silhouette	.279 ± .095	.600 ± .205	.088 ± .034	.505 ± .233
Spectral Clustering - Calinkski - Harabasz	.106 ± .027	.157 ± .425	.059 ± .042	.288 ± .215
Spectral Clustering - Eigengap	.064 ± .038	.395 ± .207	.054 ± .050	.126 ± .232

APD vs Custering-based approach

Dataset	Best WSI -> LSCD	APD
German	.873 ± .141	.760 ± .121
Spanish	.590 ± .260	.653 ± .250
English	.600 ± .205	.638 ± .202

Finding

Main Finding

- WSI optimization transfers to LSCD across all three datasets, despite differences in absolute performance.

However, sense-based clustering does not consistently outperform APD:

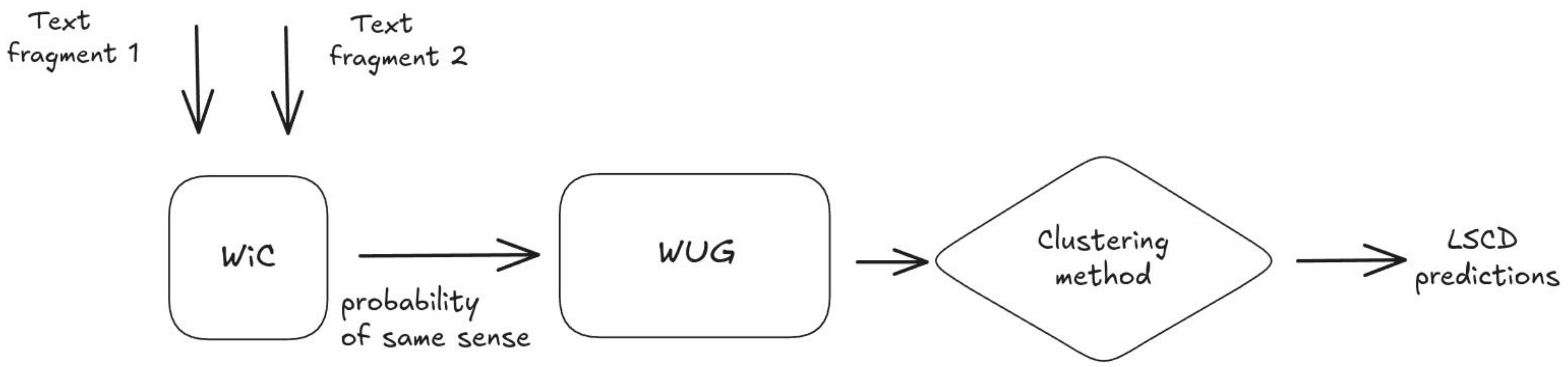
- German: Clustering > APD
- Spanish & English: APD > Clustering

Finding

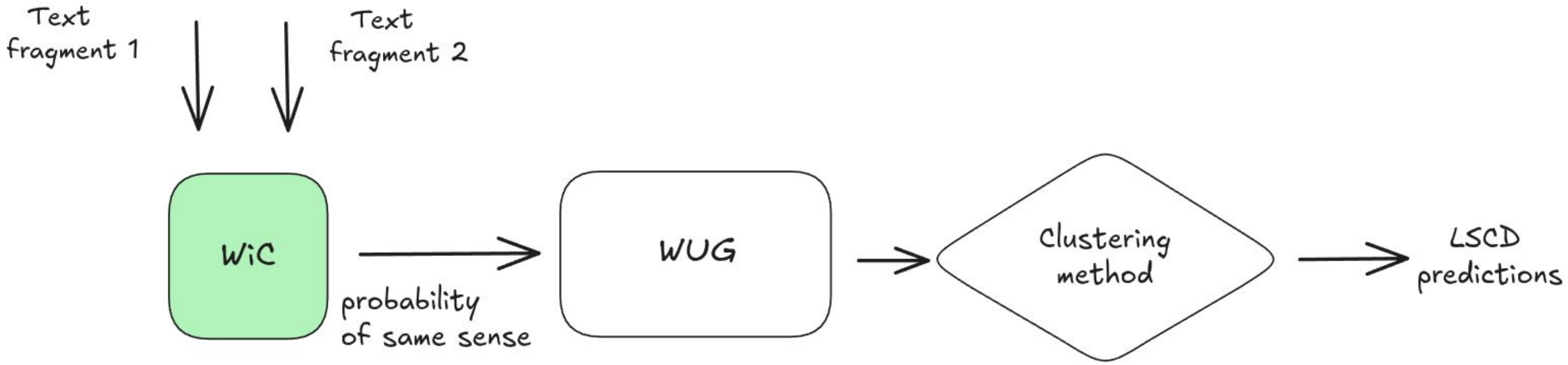
Hyperparameter analysis

- No single configuration was optimal across datasets and clustering methods.
- Hyperparameter effects were dataset- and method-dependent.

LSCD pipeline



LSCD pipeline



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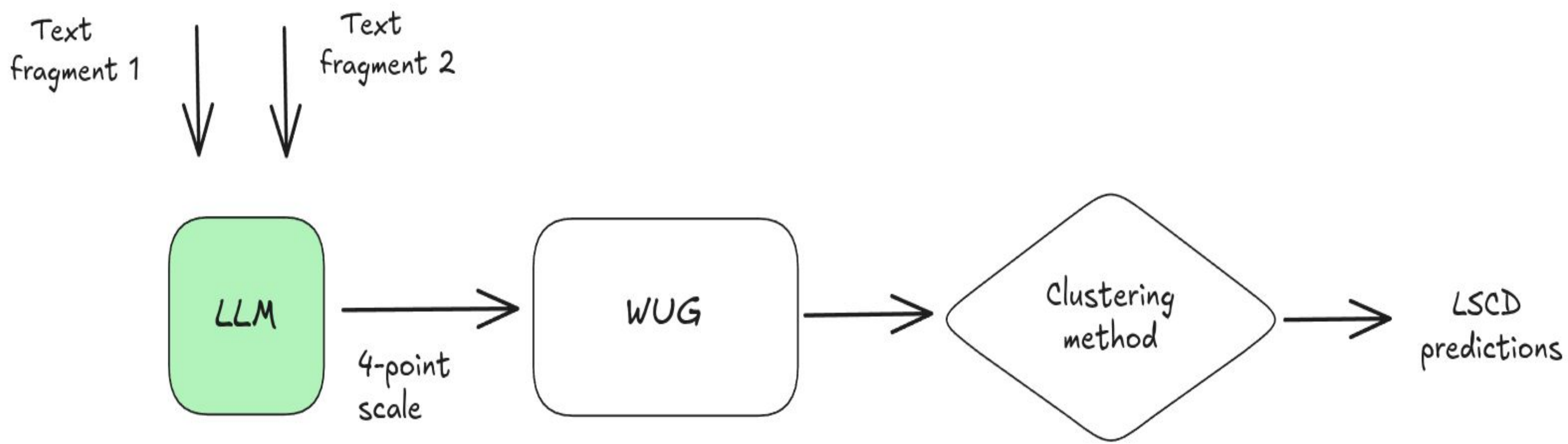

Research question

Can Large Language Models compete with specialized models in LSCD?

- Can automatically optimized prompts yield better results for the LSCD task than manually crafted prompts designed through prompt engineering?

- Can LLMs solve the Graded Change LSCD task well? Can these results surpass the WiC models reported as state-of-the-art?

- Can LLMs outperform state-of-the-art LSCD models at the annotation level?

Hypotheses

- Automatically optimized prompts improve the quality of LLM-generated semantic proximity judgments and, consequently, LSCD performance.

- LLMs can generate semantic proximity judgments that are competitive with those produced by specialized WiC models.

Metrics

- SPR_WiC - Spearman correlation between the annotations provided by human annotators and models.

- SPR_LSCD - Spearman correlation between gold graded change scores and model predictions

Models

- LLMs
  - Llama 3.1 (8B)
  - Mixtral 8x7B (Jiang et al., 2024)
  - Llama 3.3 (70B)
- WiC models
  - DeepMistake (Arefyev et al., 2021)
    - SOTA for Russian and Spanish datasets
    - MCL, enMCL, MCL-> es (fine-tuned on various WiC datasets across multiple languages)
  - XL-LEXEME (Casotti et al., 2023)
    - SOTA for English, Swedish, German datasets

Clustering methods

Spectral clustering (von Luxburg, 2007)

Agglomerative clustering (Jain and Dubes,1988)

Weighted Stochastic Block Model (Peixoto, 2019)

Clustering methods

<table>
<tr><th>Clustering method</th><th>Parameter</th><th>Values</th></tr>
<tr><td>WSBM</td><td>distribution</td><td>[binomial, poisson, discrete, normal, exponential]</td></tr>
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<tr><td>Agglomerative Clustering</td></tr>
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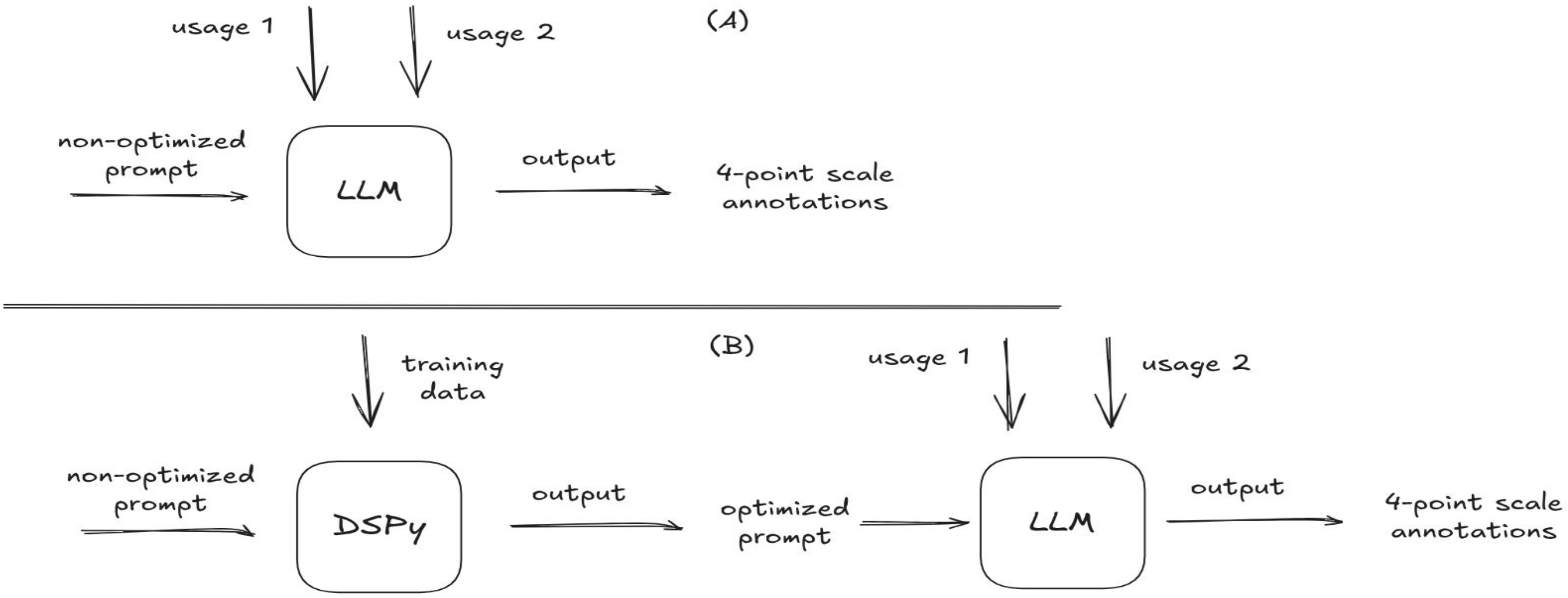
Experimental Setup - LLMs



Prompts

- optimized prompt from Yadav et al. (2024)
    - translate the prompt into Spanish
    - extend the prompt using samples from the DURel framework (Schlechtweg et al., 2018)

- optimized prompts further using DSPy framework (Khattab et al., 2023)

Prompt optimization

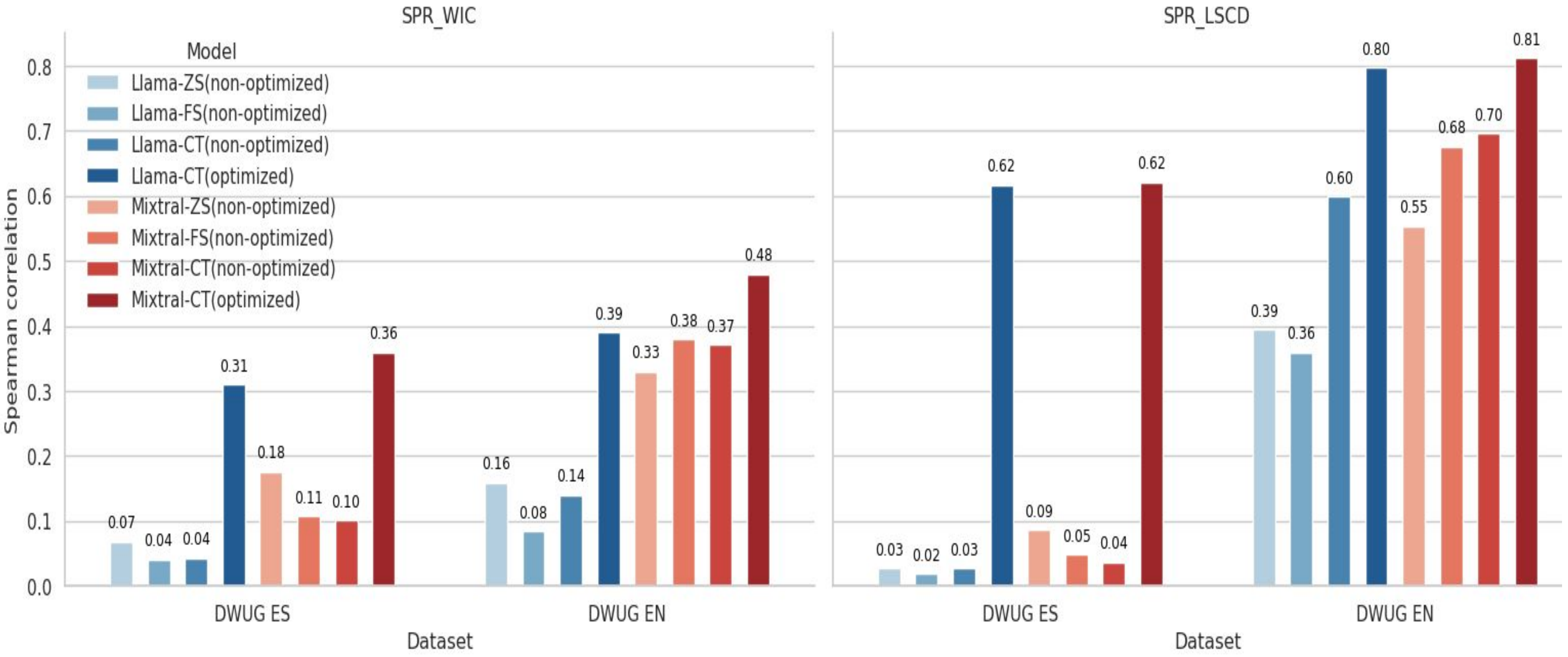
Accuracy

Dataset - prompt	Llama 3.1		Mixtral		Llama 3.3	
	NOP	OP	NOP	OP	NOP	OP
DWUG ES - PrS	26.8	29.5	32.6	31.22	32.33	39.77
DWUG ES - PrE	26.5	35.5	33.7	34.80	40.88	46.33
DWUG EN - PrS	25.75	31.0	32.8	40.75	37.25	45.5
DWUG ES - PrE	28.75	37.25	33.25	38.75	35.75	49.25

NOP - non optimized prompt, OP - optimized prompt

PrE - prompt written in English, PrS - prompt written in Spanish

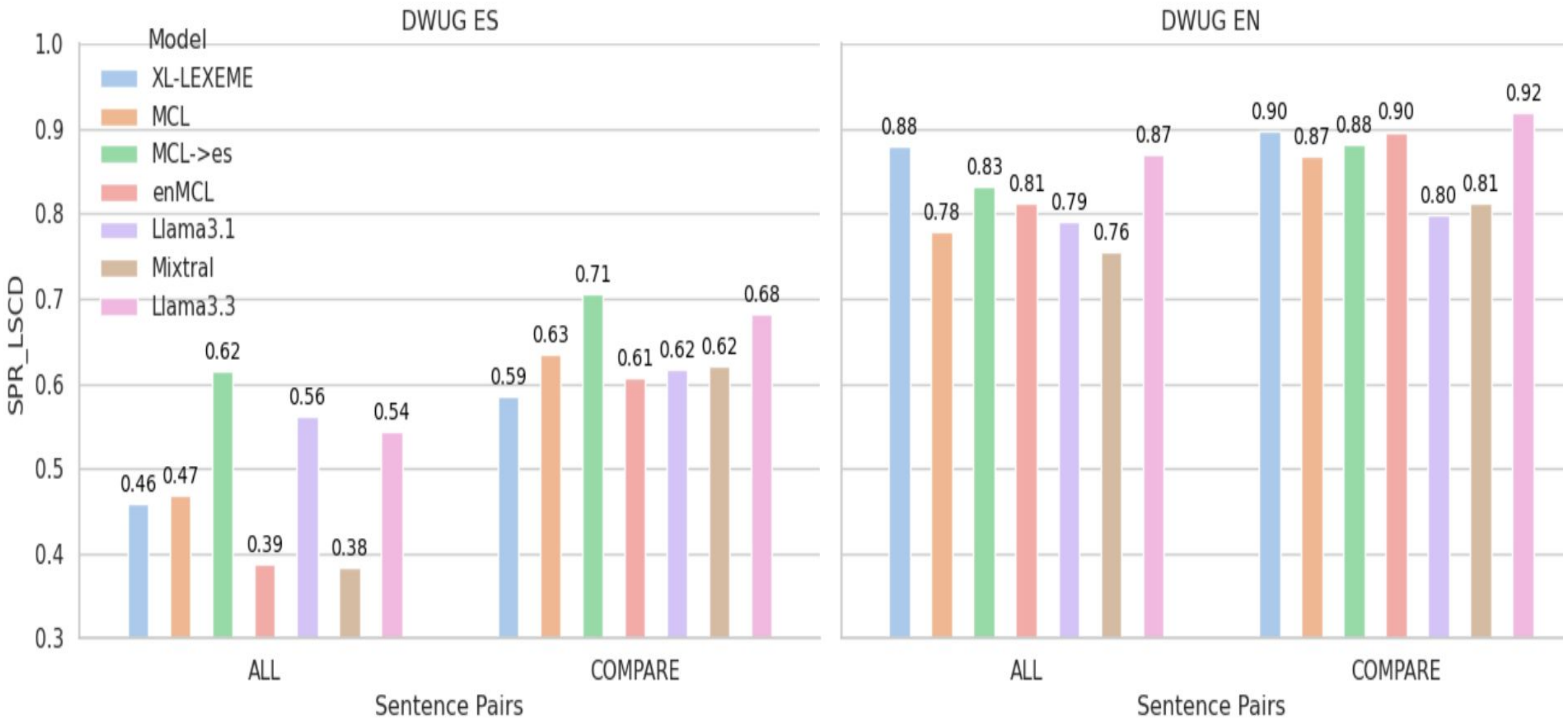
Non-optimized Prompts vs. Optimized Prompts



RQ2

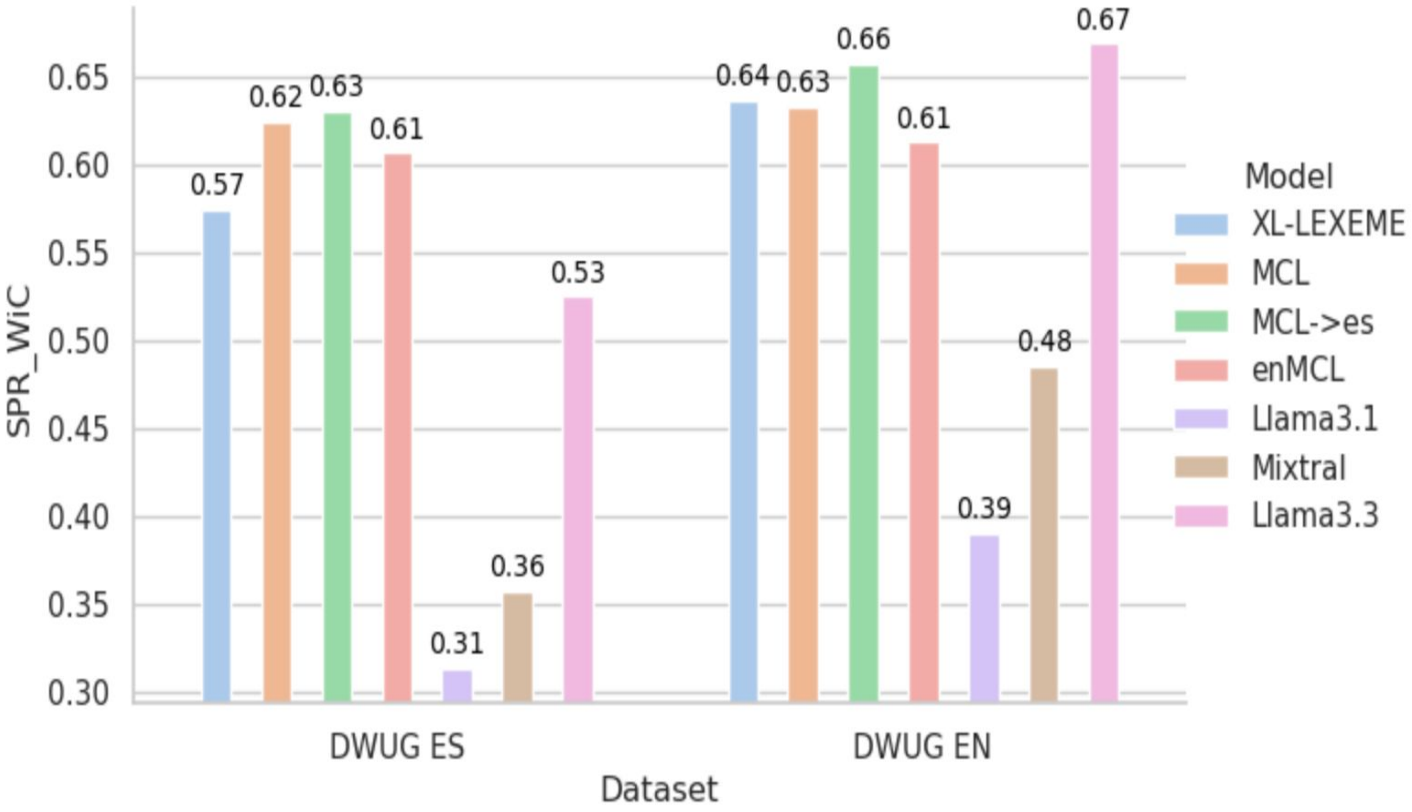
- Can LLMs solve the Graded Change LSCD task well? Can these results surpass the WiC models reported as state-of-the-art?

Specialized WiC models vs LLMs in SPR_LSCD



- Can LLMs outperform state-of-the-art LSCD models at the annotation level?

Specialized WiC models vs LLMs in SPR_WiC



Findings

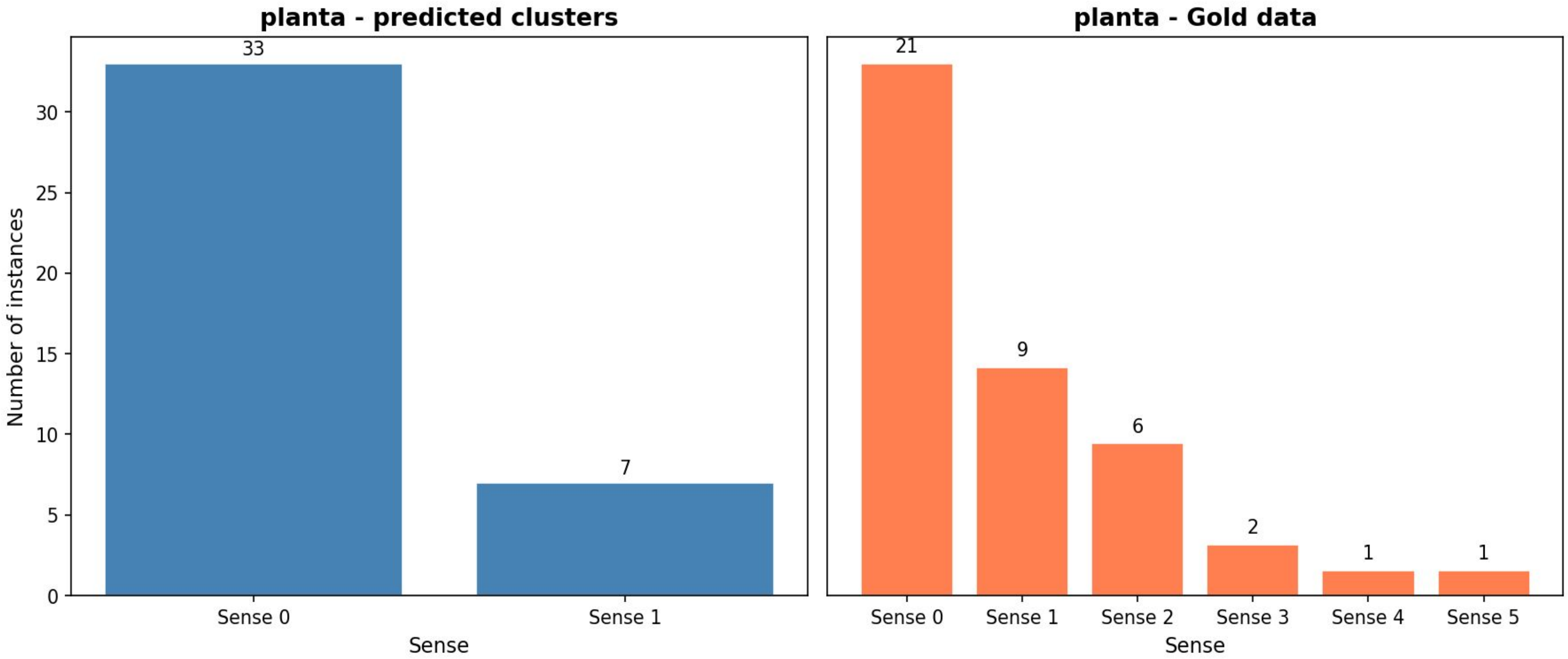
- recent prompt optimization techniques are crucial for achieving better results on the Graded Change LSCD task, as demonstrated by the performance of Llama3.3:70B

- medium-sized LLMs such as Mixtral:8x7B and Llama3.1:8B still underperform compared to smaller and faster specialized LSCD models in both the DWUG EN and DWUG ES datasets
    - in addition to optimization techniques, the size of the model also significantly influences the results

Qualitative analysis

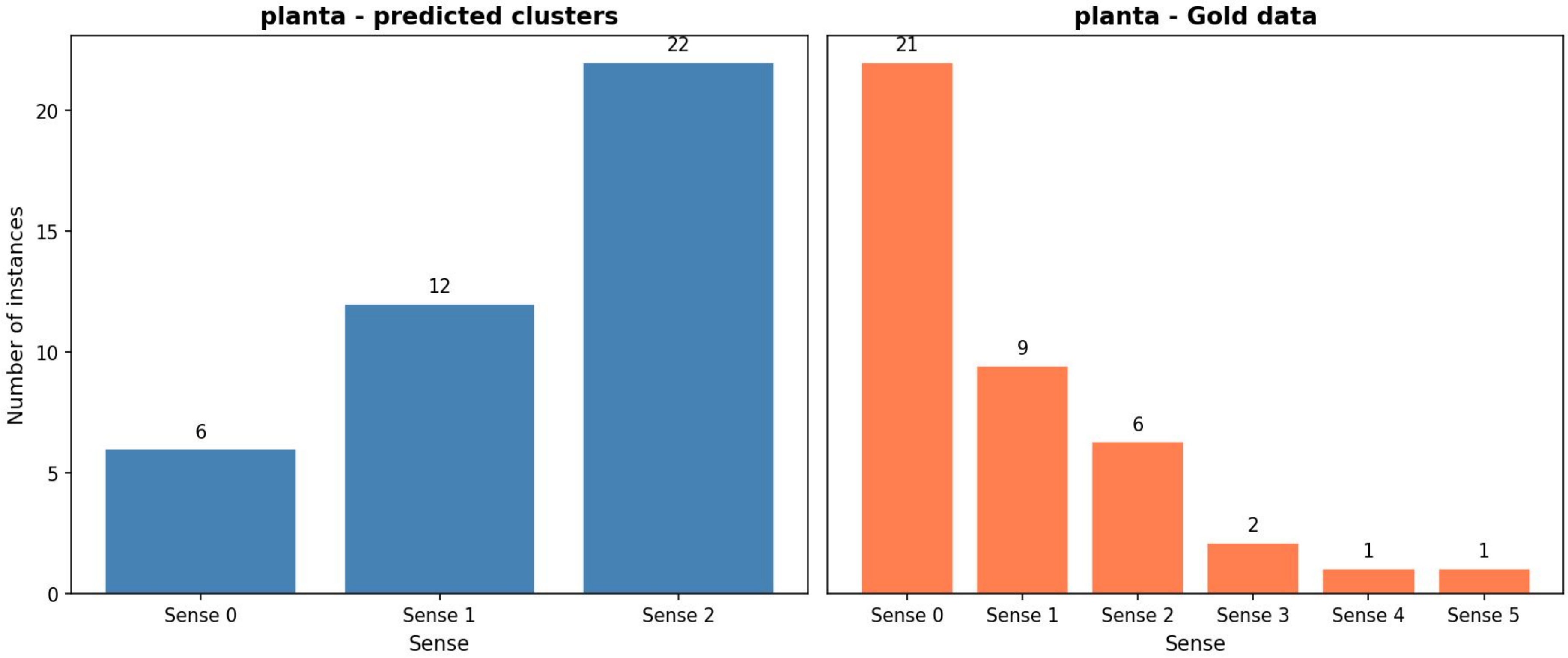
- Quantitative evaluation identifies which model performs best
- It does not explain how the semantic change is represented
- We therefore inspect the induced senses

Qualitative analysis



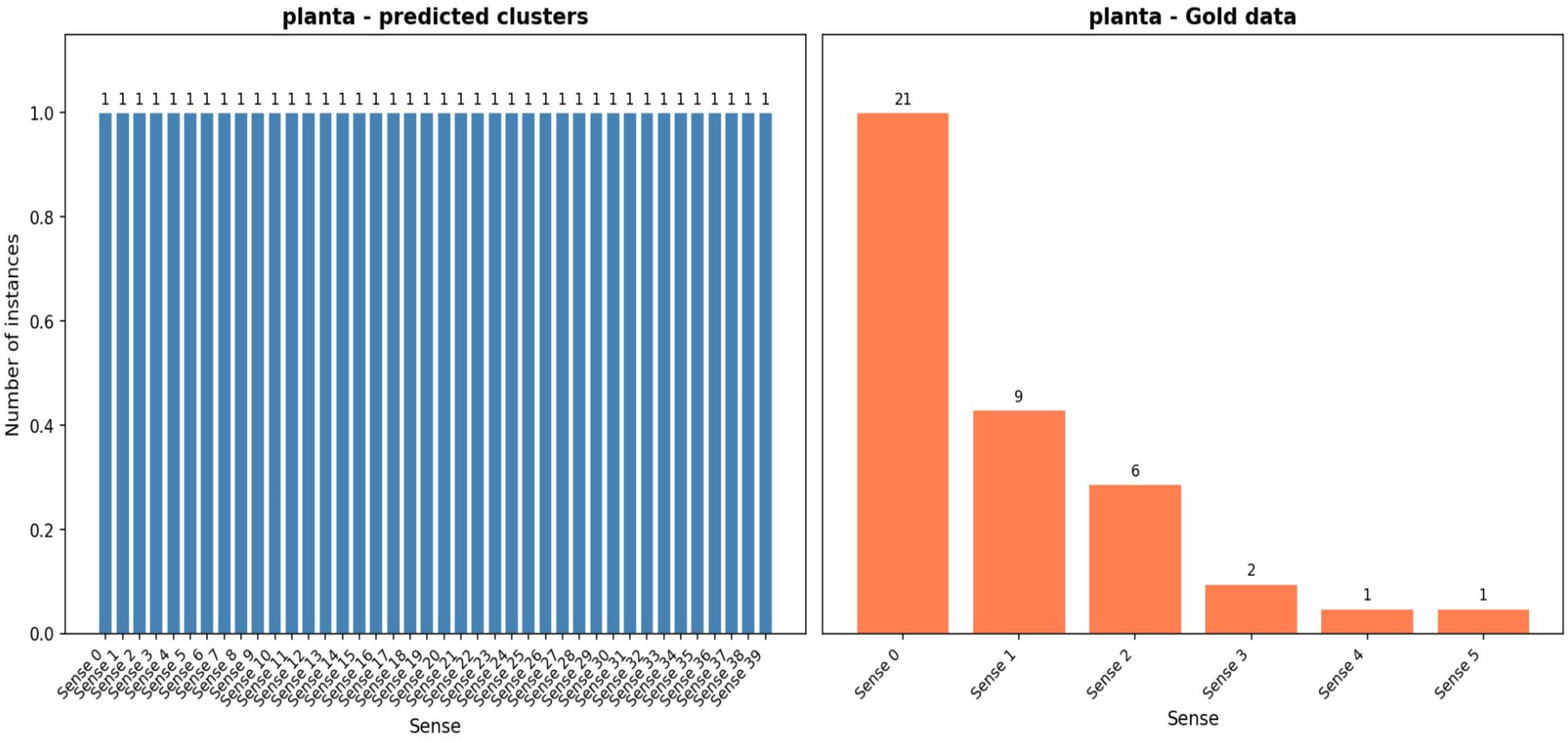
Spectral Clustering + Llama3.3:70B annotations

Qualitative analysis



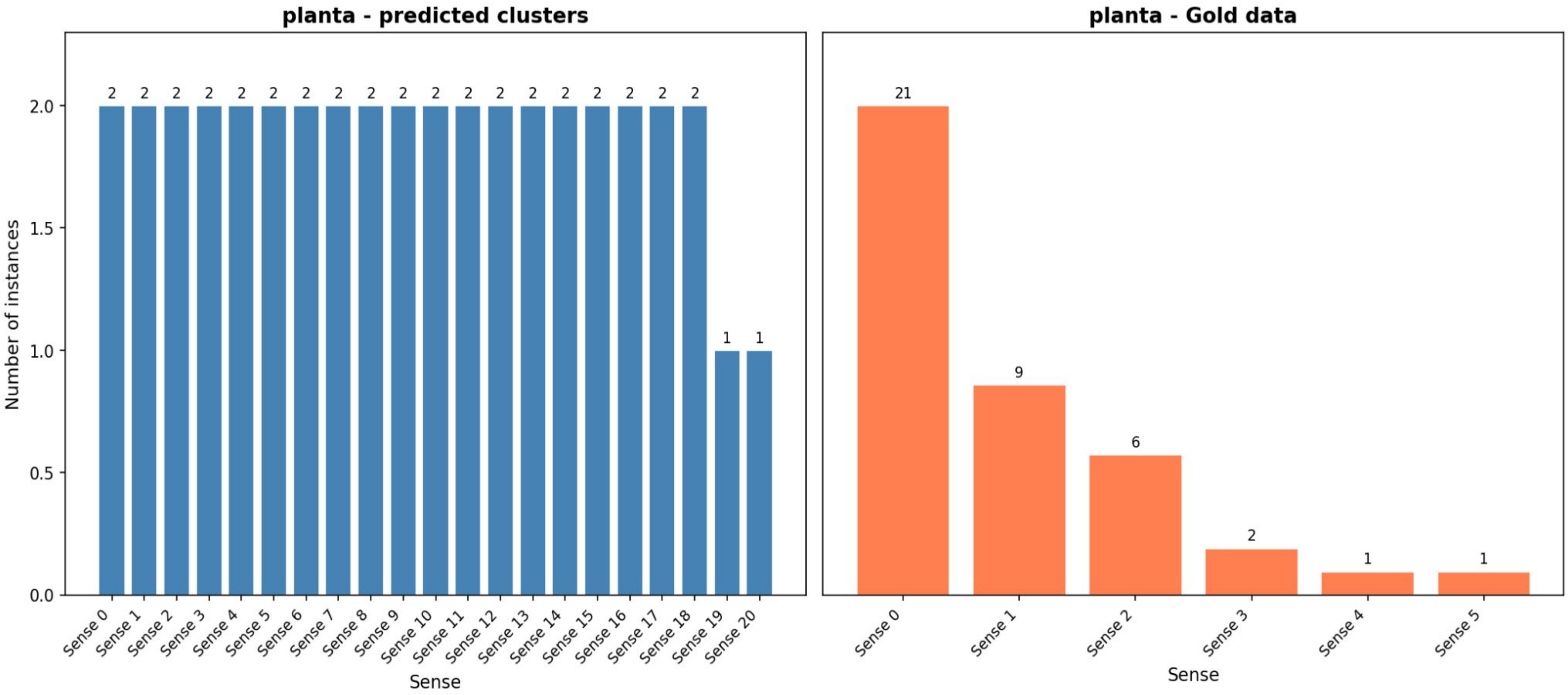
Spectral Clustering + MCL->es annotations

Qualitative analysis



WSBM + Llama3.3:70B annotations

Qualitative analysis



WSBM + MCL->es annotations

Findings - Qualitative analysis

- Different clustering methods induce substantially different sense inventories.

- Models with comparable LSCD performance may recover different semantic structures.

- Graded semantic change scores alone do not reveal the quality of the induced sense inventories.

DeepMistake model

XL-LEXEME model

Motivation

- Sense-aware LSCD methods first induce word senses and then compare their distributions across time.
- The quality of the induced sense structure may therefore directly affect semantic change detection.
- However, the relationship between WSI performance and LSCD performance has not been systematically established.